

An Overview of OpenShift



OpenShift provides all the resources needed to develop, host and scale Web applications in the public or private cloud. Available on GitHub, it has built-in support for various Web application frameworks. This article provides a tutorial on installing OpenShift and applications like WordPress, Jenkins and Tomcat through it.

Cloud computing is an innovative paradigm that delivers services on demand with a utility based billing model. NIST (National Institute of Standards and Technology, under the US Department of Commerce) has compartmentalised cloud computing into three service models (SaaS, PaaS and IaaS) and four deployment models—public clouds, private clouds, hybrid clouds and community clouds. Cloud computing is reshaping the computing and Internet landscape. With breakthroughs made in appropriate service and business models, it will certainly expand its role as the backbone for IT services. Cloud computing provides elastic and unlimited resources, which can be efficiently utilised during periods of peak loads and normal loads, with a pay per use pricing model.

As already stated, the cloud model covers Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service (IaaS). In PaaS, the user can deploy applications on the cloud infrastructure using supported programming languages (Java, PHP, Ruby and

.NET) and the platforms/tools provided by PaaS providers like Web servers, application servers and databases. Since PaaS providers manage the underlying infrastructure resources such as operating systems, virtual servers, networks, Web servers, application servers, databases, backup, disaster recovery, etc, users can focus on the most significant aspect of their business, which is the application itself, rather than worrying about managing resources, platforms and versions.

PaaS includes not only the deployment environment but also repositories such as the build environment, the testing environment, performance management, mail services, log services, database services, big data services, search services, enterprise messaging services, application performance management for modern application architectures, code inspection services and so on. OpenShift is a PaaS from Red Hat. OpenShift Origin is an open source version available on Git.

Red Hat OpenShift is a PaaS with Apache License 2.0. OpenShift Enterprise is a private cloud version.